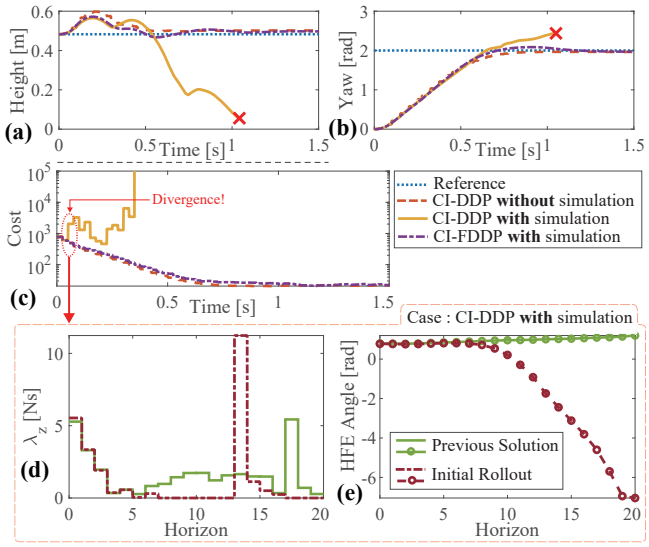

Figure 6. Comparison of Contact-implicit MPC performance under the three cases: 1) CI-DDP **without** simulation. 2) CI-DDP **with** simulation. 3) CI-FDDP **with** simulation. (a) and (b) represent the resulting height and yaw, respectively, while (c) indicate the cost associated with each sequential MPC problem. In the "CI-DDP **with** simulation" scenario, the red cross 'x' marks in (a) and (b) denote failure due to body contact with the ground, while initial divergence is marked with red dots in (c). For this instance of divergence, the deviations of the state trajectory from the previous MPC solution and the current initial roll-out trajectory are depicted in (d)-(e): (d) RF leg's normal contact impulse, and (e) RF leg's HFE angle.

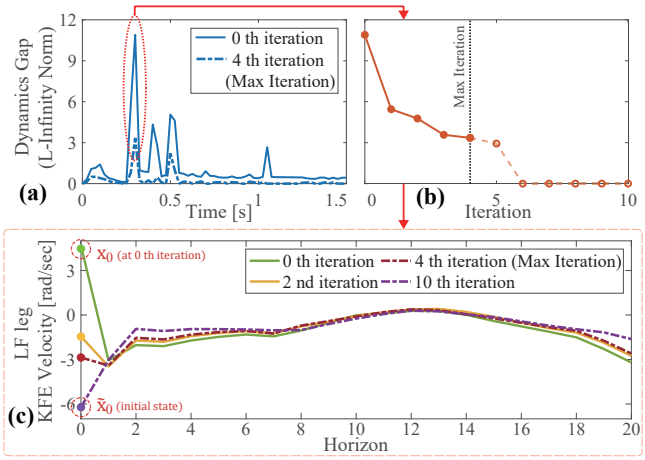

Figure 7. The detailed description for the case of CI-FDDP for the yaw turning task, shown in Figure 6. (a) denotes the dynamics gap at the initial and final iteration of MPC problems at each time step, with a maximum iteration set to 4. (b) illustrates the gap reduction process across iterations for the instance with the largest gap, marked by a red circle in (a). (c) denotes the LF leg KFE velocity trajectory along the horizon at each iteration.

6.1 Single Shooting in Contact-implicit MPC

The results in Figure 6 demonstrate the initialization issue of the single shooting scheme stemming from initial state discrepancies. The contact-implicit MPC performance is analyzed under the three cases: 1) CI-DDP **without** simulation. In this scenario, the initial state, $\tilde{\mathbf{x}}_0$, is directly taken as $\mathbf{x}_0$. This ensures that there is no initial state discrepancy, in line with solving a trajectory optimization problem in a receding horizon fashion. 2) CI-DDP **with** simulation. 3) Contact-implicit FDDP (CI-FDDP) **with** simulation. For 2) and 3) cases, RaiSim simulator (Hwangbo et al. 2018) with 1 ms integration time step is used, and the MPC runs at 40 Hz. In these cases, the initial state discrepancies are inevitable, primarily due to the first-order discretization with a relatively large time step of $dt = 25$ ms (Manchester et al. 2019). The reference body yaw angle is set as 2.0 rad, inducing a turning motion.

In the case of "CI-DDP **without** simulation", the contact-implicit MPC using a single shooting method generates multi-contact motion, achieving the desired configuration, as depicted in Figure 6 (a)-(c). However, the case of "CI-DDP **with** simulation" exhibits divergence in the early stages of the overall motion, as shown in Figure 6.

In CI-DDP cases, initialization is achieved by a roll-out of the torque trajectory. However, within the hard contact model, a slight height difference of foot can lead to unplanned contact loss. In Figure 6 (d), the right front (RF) leg is shown to lose ground contact prematurely. Instead of maintaining contact throughout the entire prediction horizon (N=20), the foot loses contact after just 5 time steps. Consequently, this alters the torque deployment, originally planned for ground reaction force (GRF), accelerating the joint velocity and eventually resulting in the divergence of joint position, observed in Figure 6 (e). Such changes also pose challenges in maintaining balance due to insufficient GRF. As a result, the initialized trajectory deviates significantly from the solution trajectory of the previous step.

6.2 Multiple Shooting for Contact-implicit MPC

As evident in "CI-FDDP **with** simulation" of the yaw turning task depicted in Figure 6 (a)-(c), the CI-FDDP accomplishes the task despite initial state discrepancies. At the start of motion, the first MPC problem is initialized with a feasible state-control trajectory, outlined in following subsection. Then, the dynamics gap is exclusively attributed to the initial state discrepancy, and that dynamics gap at the 0 th iteration can be expressed as follows:

$$\bar{\mathbf{f}}_0 := \tilde{\mathbf{x}}_0 \ominus \mathbf{x}_0 \neq \mathbf{0},$$
$$\bar{\mathbf{f}}_{i+1} := \mathbf{f}(\mathbf{x}_i, \mathbf{u}_i) \ominus \mathbf{x}_{i+1} = \mathbf{0}, \quad \forall i = \{0,1,\ldots,N-1\}.$$

During the line search procedure in the forward pass, the initial roll-out state $\hat{\mathbf{x}}_0$ is determined along the step size α as follows: $\hat{\mathbf{x}}_0 := \tilde{\mathbf{x}}_0 \oplus (\alpha - 1)\bar{\mathbf{f}}_0$, where it denotes the interpolation point lying between $\tilde{\mathbf{x}}_0$ and $\mathbf{x}_0$. If the roll-out trajectory is approved according to the acceptance condition, then the roll-out state $\hat{\mathbf{x}}_i$ is adopted as shooting state $\mathbf{x}_i$, and this entire process iterates until the gap is reduced to zero.

A detailed process of gap contraction is presented in Figure 7. Figure 7 (a) shows the dynamics gap at each MPC problem's initial and final iterations across the time steps. The detailed gap reduction process is illustrated in Figure 7 (b). Furthermore, the shooting state trajectory is updated along the iterations, observed in Figure 7 (c). In each iteration, the optimizer incrementally adjusts the trajectory by pulling the 0 th shooting state $\mathbf{x}_0$ closer to the initial state $\tilde{\mathbf{x}}_0$, facilitated by the progressive modification using step size α, as shown in Figure 7 (c). Meanwhile, the other states are closely preserved to the state of the previous iteration, due to

the feedback term (gain K in equation (5)), implicitly guiding the contact modes to align with prior modes.

In conclusion, during the initialization, CI-FDDP prioritizes consistency with prior contact modes by trading off feasibility, thereby enhancing the robustness of initialization. Conversely, CI-DDP risks deviating from prior contact modes and often diverges as a result, when integrating the state trajectory to achieve feasible initialization. In practice, CI-FDDP also manages unexpected contact events (Section 7.2.5), and conducts real-robot experiments despite model imperfections and state estimation uncertainties.

6.2.1 Initial Feasible Trajectory. Prior to the initial MPC problem, we solve a trajectory optimization problem under the reference configuration with a standing posture, using the same framework. This aims to initialize the initial MPC with a *feasible* state-control trajectory. All varied motions in this paper stem from this standing motion initialization, not requiring the task-specified pre-planned trajectories.

6.2.2 Maximum Iteration. To ensure real-time computation requirements, we limit the maximum number of iterations of the CI-FDDP as 4. The optimizer requires minimal iterations to reduce the dynamics gap and explore contact modes in each iteration. As evident in Figure 7 (a), despite the optimizer's inability to eliminate all gaps in every instance due to the limitation of maximum iteration, the subsequent MPC problems could compensate it with fast recomputation setting.

6.3 Feedback Control

Similar to Youm et al. (2023), we proposed a PD control that combines feed-forward torques with feedback actions:

$$\mathbf{p}_{\text{target}} = \mathbf{u}_0^*/K_p + \mathbf{p}_{\text{des}},$$
$$\mathbf{u}_{\text{cmd}} = \mathbf{K}_p(\mathbf{p}_{\text{target}} - \mathbf{p}) + \mathbf{K}_d(\dot{\mathbf{p}}_{\text{des}} - \dot{\mathbf{p}}).$$

Here, $\mathbf{p}_{\text{des}}$ and $\dot{\mathbf{p}}_{\text{des}}$ are chosen based on the next time step's joint state from the solution trajectory of the CI-FDDP. If the current joint position and velocity, $\mathbf{p}$ and $\dot{\mathbf{p}}$, are close to $\mathbf{p}_{\text{des}}$ and $\dot{\mathbf{p}}_{\text{des}}$, respectively, $\mathbf{u}_{\text{cmd}}$ aligns with the feed-forward torque $\mathbf{u}_0^*$. Feedback actions increase as they deviate from $\mathbf{p}_{\text{des}}$ and $\dot{\mathbf{p}}_{\text{des}}$.

In our target scenario of dynamic motions, such as front-leg rearing, frequent contact mismatches occur due to rapid changes in contact mode. PD control could robustly handle these unexpected contact events. For example, in the late (or missing) contact cases, the PD control could decelerate an accelerated leg, and in the early contact cases, it could put more torques for the additional ground reaction forces. In our work, this straightforward implementation of PD control is sufficient to execute the motions planned by the MPC [4].

7 Results

We present the results of our framework using quadruped robots. First, we analyze the effect of the relaxation variable ρ using a 2D planar model. Next, we showcase multi-contact motions for various desired configurations in 3D simulations, and compare our framework with MuJoCo MPC (Howell et al. 2022). Lastly, we validate the contact-implicit MPC framework with hardware experiments, illustrating trot and front-leg rearing motions.

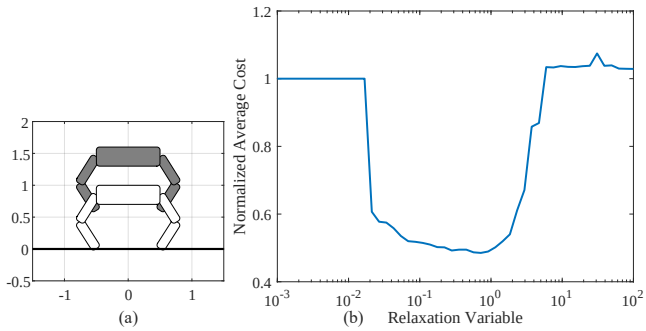

Figure 8. Relation between relaxation variable and average cost for jumping up task : (a) 2D environment, grey robot for the reference trajectory and white robot for actual robot. (b) Result of the average cost according to the relaxation variable.

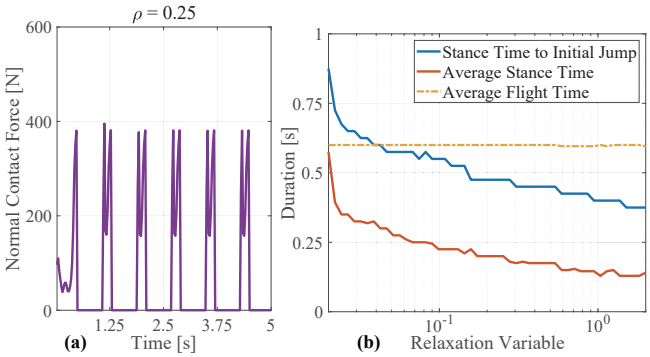

Figure 9. Periodic jumping motion analysis: (a) Contact impulse for 0.25 relaxation variable, indicating periodic jumps. (b) Average stance time versus relaxation variable.

7.1 Effect of the Relaxation Variable

We investigate the impact of the relaxation variable ρ, which allows the gradient to navigate motions that involve breaking contact. We conducted a 2D simulation with a planar quadruped robot, following the settings from our earlier work (Kim et al. 2022). The framework solely employed a regulating cost with a fixed reference configuration. Figure 8 (a) illustrates the 2D setup and Figure 8 (b) maps the average cost against the relaxation variable.

In this scenario, with a reference trajectory set significantly above the robot, periodic jumping results in a lower cost than standing still. Without relaxation, the motion remains standing, aligning with the initialized trajectory where all feet are in contact, until the relaxation variable exceeds 0.02. Beyond this value, the robot exhibits a periodic jumping motion, achieving a lower cost than standing. However, when the relaxation variable exceeds 6, the robot prematurely attempts to jump without accumulating sufficient contact impulse. This leads to in-place bouncing, increasing the cost due to unnecessary movement, as illustrated in Figure 8 (b). All the discovered motions are in Extension 3.

Notably, there are no modifications to the cost function, such as weight adjustments or time-dependent costs. The smooth gradient (through relaxation) enables the emergence of the jumping motion, leading to a lower-cost solution within the same optimal control problem.

We further investigate the effect of relaxation variable ρ within the scope of [0.02, 2], exhibiting periodic jumping motions. Figure 9 (a) depicts the contact impulses for